

US Sector Allocation by Rate, Inflation & Business Cycle

Independent research study | Mo Rahmani

Sample: 1932-06-30 to 2023-09-30 (1096 months)

Executive summary

Across roughly 90 years of US equity industry data, the clearest regime edges are: overweight Energy when headline CPI is high and/or long rates are rising; overweight NonDurables and Health Care when industrial production is contracting; overweight Durables and Business Equipment when inflation is low and long rates are falling. A simple top-3 Sharpe sleeve adds 3-9 percentage points annualized vs equal-weight in high-inflation and contraction cells, but little in the common expansion/mid-inflation state.

Data & methodology

Sources:

- Ken French 12 Industry Portfolios (value-weighted monthly)
- Shiller long interest rate GS10 proxy (ie_data)
- Headline CPI: Shiller CPI, extended with FRED CPIAUCNS when available
- FRED INDPRO (industrial production, 1919+)

Regime definitions:

- rate_level: High/Low vs expanding median of Shiller long rate (min 60m)
- rate_dir: Rising/Falling = 12m change in Shiller long rate ≥ 0
- infl_level: Low/Mid/High = expanding tertiles of CPI YoY (min 60m)
- infl_dir: Rising/Falling = 12m change in CPI YoY ≥ 0
- cycle: Expansion if INDPRO YoY ≥ 0 else Contraction

Caveats:

- French industries are SIC-based, not modern GICS; BusEq ~ tech/equipment, not pure XLK.
- In-sample conditional means are not a live trading system; use walk-forward before allocating capital.
- Regime labels use expanding history to limit look-ahead on level thresholds.
- Shiller long rate is a long government yield proxy, not the Fed funds rate.

Business cycle (INDPRO YoY)

Expansion (820 months, 74.8% of sample)

Best: Energy (15.07% ann, Sharpe 0.77), Telecom (10.64% ann, Sharpe 0.73), Consumer NonDurables (9.89% ann, Sharpe 0.68)

Worst: Consumer Durables (11.92% ann, Sharpe 0.5), Other (mines/trans/hotels) (10.06% ann, Sharpe 0.53), Utilities (9.42% ann, Sharpe 0.59)

Contraction (276 months, 25.2% of sample)

Best: Consumer NonDurables (19.87% ann, Sharpe 1.14), Retail / Shops (23.54% ann, Sharpe 1.06), Health Care (21.54% ann, Sharpe 1.0)

Worst: Energy (11.25% ann, Sharpe 0.42), Telecom (9.3% ann, Sharpe 0.51), Other (mines/trans/hotels) (18.31% ann, Sharpe 0.63)

Long-rate direction (Shiller GS10, 12m change)

Rising long rates (559 months, 51.0% of sample)

Best: Energy (13.39% ann, Sharpe 0.69), Consumer NonDurables (7.72% ann, Sharpe 0.56), Business Equipment (10.93% ann, Sharpe 0.55)

Worst: Consumer Durables (7.42% ann, Sharpe 0.35), Telecom (5.21% ann, Sharpe 0.38), Other (mines/trans/hotels) (6.77%

ann, Sharpe 0.38)

Falling long rates (537 months, 49.0% of sample)

Best: Consumer NonDurables (17.28% ann, Sharpe 1.03), Retail / Shops (20.97% ann, Sharpe 1.0), Health Care (18.44% ann, Sharpe 0.94)

Worst: Energy (14.85% ann, Sharpe 0.63), Business Equipment (19.22% ann, Sharpe 0.69), Utilities (14.36% ann, Sharpe 0.7)

Inflation level (headline CPI YoY tertiles)

High inflation (459 months, 41.9% of sample)

Best: Energy (12.44% ann, Sharpe 0.58), Health Care (9.88% ann, Sharpe 0.54), Consumer NonDurables (8.14% ann, Sharpe 0.52)

Worst: Consumer Durables (4.8% ann, Sharpe 0.2), Other (mines/trans/hotels) (5.79% ann, Sharpe 0.27), Manufacturing (6.52% ann, Sharpe 0.32)

Mid inflation (483 months, 44.1% of sample)

Best: Consumer NonDurables (14.36% ann, Sharpe 1.09), Retail / Shops (14.93% ann, Sharpe 0.95), Utilities (13.23% ann, Sharpe 0.93)

Worst: Business Equipment (16.23% ann, Sharpe 0.73), Consumer Durables (15.95% ann, Sharpe 0.74), Other (mines/trans/hotels) (14.0% ann, Sharpe 0.78)

Low inflation (154 months, 14.1% of sample)

Best: Consumer Durables (45.26% ann, Sharpe 1.03), Chemicals (26.62% ann, Sharpe 1.02), Retail / Shops (24.4% ann, Sharpe 0.99)

Worst: Utilities (14.08% ann, Sharpe 0.59), Energy (17.98% ann, Sharpe 0.6), Finance (25.25% ann, Sharpe 0.74)

Allocation playbook

Rising long rates

Overweight: Energy (+5.1pp excess vs EW)

Underweight: Telecom, Durables, Retail

Why: Higher discount rates hurt duration assets; energy co-moves with inflation/hiking cycles.

Falling long rates

Overweight: Durables, Retail, Health (Sharpe)

Underweight: Energy, Utilities (relative)

Why: Lower yields re-rate discretionary/growth; inflation hedge less needed.

High CPI

Overweight: Energy (+4.7pp), Health, NonDurables

Underweight: Durables, cyclicals

Why: Commodity pass-through and staples pricing power.

Low CPI

Overweight: Durables, Chemicals, Bus. Equipment

Underweight: Utilities, Energy, Telecom

Why: Risk-on / easier financial conditions (thin sample: 154 months).

INDPRO expansion

Overweight: Energy, Bus. Equipment, Manufacturing

Underweight: Utilities, NonDurables (relative)

Why: Capex and commodity demand lift cyclicals.

INDPRO contraction

Overweight: NonDurables, Health (Sharpe)

Underweight: Energy (-7.3pp excess)

Why: Defensives on risk adjustment; energy lags when activity falls.

Top-3 Sharpe sleeve vs equal-weight (Cycle x Inflation)

Regime cell	N	Top 3 industries	Sharpe	Edge (pp)
Expansion / MidInfl	403	Consumer NonDurables, Retail / Shops, Telecom	0.13	-0.21
Expansion / HighInfl	343	Energy, Business Equipment, Telecom	0.57	+3.29
Contraction / HighInfl	116	Consumer NonDurables, Retail / Shops, Health Care	1.02	+8.61
Contraction / LowInfl	80	Consumer NonDurables, Consumer Durables, Retail / Shops	1.55	+6.94
Contraction / MidInfl	80	Health Care, Retail / Shops, Finance	1.19	+4.15
Expansion / LowInfl	74	Health Care, Telecom, Chemicals	1.31	+1.62

Historical episodes

Great Depression

Avg long rate 3.42% | CPI YoY -6.76% | INDPRO YoY -15.94%

Best: Consumer NonDurables (-16.33%), Health Care (-20.1%), Telecom (-21.06%)

Worst: Business Equipment (-39.9%), Manufacturing (-37.74%), Finance (-36.47%)

Industrial collapse, deflation, bank failures - cyclicals and finance crushed; relative resilience in necessities.

WWII / postwar boom

Avg long rate 2.41% | CPI YoY 5.22% | INDPRO YoY 7.42%

Best: Finance (32.26%), Consumer Durables (31.69%), Other (mines/trans/hotels) (31.27%)

Worst: Chemicals (13.93%), Telecom (17.07%), Health Care (21.76%)

War production and controlled economy; manufacturing and durables demand; financial repression on rates.

1970s stagflation

Avg long rate 8.96% | CPI YoY 9.05% | INDPRO YoY 2.45%

Best: Energy (15.35%), Telecom (10.08%), Other (mines/trans/hotels) (9.9%)

Worst: Consumer Durables (3.58%), Retail / Shops (4.06%), Chemicals (5.25%)

Oil shocks + high CPI + rising long rates. Energy and hard assets thrived; long-duration growth and bond proxies lagged.

Volcker disinflation

Avg long rate 12.61% | CPI YoY 10.23% | INDPRO YoY -1.95%

Best: Retail / Shops (25.15%), Consumer NonDurables (21.71%), Health Care (21.14%)

Worst: Chemicals (10.95%), Energy (11.02%), Manufacturing (12.1%)

Violent rate hikes crushed inflation and activity; deep double-dip recession - rate-sensitive cyclicals and finance under pressure until rates peaked.

Great Moderation / bull market

Avg long rate 7.9% | CPI YoY 3.27% | INDPRO YoY 3.72%

Best: Telecom (20.55%), Business Equipment (19.45%), Health Care (18.62%)

Worst: Utilities (12.86%), Other (mines/trans/hotels) (12.97%), Consumer Durables (14.53%)

Falling inflation and falling long rates - duration assets and growth/tech (BusEq) compounded; equity risk premium compressed.

Dot-com boom & bust

Avg long rate 5.74% | CPI YoY 2.45% | INDPRO YoY 3.49%

Best: Finance (17.46%), Health Care (15.77%), Business Equipment (14.5%)

Worst: Other (mines/trans/hotels) (4.42%), Telecom (6.79%), Consumer Durables (8.37%)

Business Equipment / tech extreme boom then crash; defensives and staples relatively better in the bust.

Global Financial Crisis

Avg long rate 3.61% | CPI YoY 3.22% | INDPRO YoY -4.12%

Best: Health Care (-16.97%), Retail / Shops (-18.79%), Consumer NonDurables (-19.59%)

Worst: Consumer Durables (-67.93%), Finance (-63.67%), Manufacturing (-48.51%)

Credit collapse - Finance catastrophic; Energy and cyclicals sold off with INDPRO; Utilities/Health relatively defensive.

Post-GFC QE / low-rate era

Avg long rate 2.47% | CPI YoY 1.62% | INDPRO YoY 0.84%

Best: Business Equipment (19.63%), Consumer Durables (19.47%), Manufacturing (18.31%)

Worst: Energy (7.01%), Utilities (12.74%), Consumer NonDurables (14.15%)

ZIRP + QE: falling/low long rates supported duration (BusEq, growth) and housing-linked cyclicals; Energy lagged in mid-2010s oil bust.

COVID shock & rebound

Avg long rate 1.14% | CPI YoY 2.99% | INDPRO YoY -1.13%

Best: Consumer Durables (80.45%), Business Equipment (34.6%), Retail / Shops (26.81%)

Worst: Telecom (5.18%), Utilities (6.87%), Consumer NonDurables (12.41%)

Sudden INDPRO collapse then fiscal/monetary surge; Tech/BusEq and discretionary rebound; Energy initially crushed then recovered.

2021-23 inflation & hiking cycle

Avg long rate 2.62% | CPI YoY 5.56% | INDPRO YoY 2.08%

Best: Energy (37.02%), Business Equipment (13.18%), Consumer Durables (11.91%)

Worst: Telecom (-6.56%), Chemicals (3.48%), Health Care (4.83%)

CPI spike + rising long rates: Energy and some value/cyclicals led early; long-duration BusEq/growth hurt in 2022, then recovered as inflation cooled.

Decade champions

Decade	CPI	Rate	Best	Worst
1930s	-0.5%	2.8%	Consumer Durables (41.1%)	Utilities (14.9%)
1940s	5.7%	2.3%	Energy (14.2%)	Telecom (5.3%)
1950s	2.1%	3.0%	Business Equipment (24.4%)	Consumer NonDurables (11.9%)
1960s	2.3%	4.7%	Health Care (13.2%)	Chemicals (4.7%)
1970s	7.1%	7.5%	Energy (14.5%)	Business Equipment (5.0%)
1980s	5.6%	10.6%	Consumer NonDurables (24.5%)	Business Equipment (10.2%)
1990s	3.0%	6.7%	Business Equipment (28.6%)	Utilities (9.0%)
2000s	2.6%	4.5%	Energy (12.7%)	Telecom (-5.6%)
2010s	1.8%	2.4%	Business Equipment (16.6%)	Energy (4.9%)
2020s	4.6%	2.2%	Consumer Durables (40.2%)	Telecom (-3.9%)

Sector fundamentals (why regimes matter)

Energy

Commodity cash flows; inflation hedge when CPI is supply-driven; lags in deep contractions.

Business Equipment

Long-duration growth/capex proxy; compounds when rates fall; hurt by rising discount rates.

Utilities / Telecom

Bond-proxy cash flows; reprice lower when long rates rise.

NonDurables / Health

Inelastic demand and pricing power; defensive Sharpe in contractions and high CPI.

Durables / Retail

Rate- and income-sensitive discretionary spend; strong in falling-rate / low-CPI regimes.

Finance

Credit-cycle leverage; catastrophic left tail in systemic crises (GFC).

Disclaimer

This document summarizes in-sample conditional statistics on Ken French SIC-based industry portfolios. It is research commentary, not investment advice. Past regime relationships may not persist. Validate with walk-forward tests before

allocating capital.